SQL Practice Questions 50-100

Advanced SQLite workbook | Customer, product, category and time-series analytics

SKILLS DEVELOPED

- **Customer and product analytics**
- **EXISTS, NOT EXISTS and conditional aggregation**
- **RANK, DENSE_RANK, ROW_NUMBER and LAG**
- **SQLite strftime() and julianday()**
- **Running totals, growth rates and comprehensive reporting**

Question-only edition | Designed for the existing employees, customers, products and orders tables

Workbook map

Questions progress from customer order analysis to multi-stage interview-level reporting.

Level	Focus	Questions
Level 8	Customer Order Analysis	Q50-Q55
Level 9	Customer Behaviour	Q56-Q65
Level 10	Product Analysis	Q66-Q75
Level 11	Category Analysis	Q76-Q85
Level 12	Date and Monthly Sales Analysis	Q86-Q95
Level 13	Advanced Assessment Questions	Q96-Q100

Recommended approach

1. Identify the required output grain.
2. Aggregate before ranking when the task asks for revenue or spending rank.
3. Use year-month values so different years are not combined.
4. Test ties, NULL values and entities with no matching orders.

LEVEL 8 Customer Order Analysis

Customer-level measures, order values and first or latest purchase identification.

Q50

Customer average order value

Calculate the average order amount for each customer.

JOIN / AVG

Q51

Above-average customer order value

Find customers whose average order value is greater than the overall average order amount.

CTE / AVG

Q52

Highest-value customer order

Find the highest-value individual order placed by each customer.

MAX / GROUP BY

Q53

Lowest-value customer order

Find the lowest-value individual order placed by each customer.

MIN / GROUP BY

Q54

First customer purchase

Find the first order placed by each customer.

ROW_NUMBER

Q55

Latest customer purchase

Find the most recent order placed by each customer.

ROW_NUMBER

LEVEL 9 Customer Behaviour

Distinct purchasing patterns, category behaviour, existence tests and customer ranking.

Q56

First and last purchase dates

Find the first and last order date for each customer.

MIN / MAX

Q57

Customer activity period

Calculate the number of days between each customer's first and last orders using julianday().

DATE FUNCTIONS

Q58

Customers purchasing multiple products

Find customers who ordered more than one distinct product.

COUNT DISTINCT

Q59

Customers purchasing multiple categories

Find customers who purchased products from more than one category.

MULTI-TABLE JOIN

Q60

Electronics customers

Find customers who purchased at least one Electronics product using EXISTS.

EXISTS

Q61

Non-Furniture customers

Find customers who never purchased a Furniture product using NOT EXISTS.

NOT EXISTS

Q62

Customers purchasing both categories

Find customers who purchased products from both Electronics and Furniture.

CONDITIONAL AGGREGATION

Q63

Customers purchasing only Electronics

Find customers who purchased Electronics products but never purchased Furniture products.

EXISTS / NOT EXISTS

Q64

Customer spending rank

Rank customers by total spending from highest to lowest.

RANK

Q65

Top three customers

Find the three customers with the highest total spending.

DENSE_RANK

LEVEL 10 Product Analysis

Product order activity, revenue ranking and contribution analysis.

Q66

Product order summary

Find order count and total revenue for each product, including products with no orders.

LEFT JOIN / COALESCE

Q67

Products never ordered

Find products that have never appeared in the orders table.

ANTI JOIN

Q68

Products ordered several times

Find products that have been ordered at least three times.

HAVING

Q69

Product average sales amount

Calculate the average order amount associated with each product.

AVG

Q70

Product revenue rank

Rank all products based on total revenue.

RANK

Q71

Product rank within category

Rank each product by total revenue within its category.

PARTITION BY

Q72

Top two products per category

Find the two highest-revenue products in each category.

DENSE_RANK

Q73

Lowest-revenue product per category
Find the product generating the lowest total revenue in each category.
RANK

Q74

Products above average revenue
Find products whose total revenue is greater than average product revenue.
CTE / SUBQUERY

Q75

Product revenue contribution
Calculate each product's percentage contribution to total revenue.
WINDOW AGGREGATE

LEVEL 11 Category Analysis

Category pricing, revenue summaries and within-category comparisons.

Q76

Category product count
Find the number of products in each category.
COUNT / GROUP BY

Q77

Category average list price
Calculate the average product price for each category.
AVG

Q78

Category price range
Find the minimum and maximum product price in each category.
MIN / MAX

Q79

Category revenue summary
Calculate order count, total revenue and average order amount for each category.
MULTIPLE AGGREGATES

Q80

Highest-revenue category
Find the category generating the highest total revenue.
CTE / MAX

Q81

Category revenue rank
Rank product categories according to total revenue.
RANK

Q82

Categories above average revenue
Find categories whose total revenue is greater than average category revenue.
CTE / AVG

Q83

Category revenue percentage
Calculate each category's percentage contribution to overall revenue.
WINDOW AGGREGATE

Q84

Most expensive product per category
Find the most expensive product in each category.
DENSE_RANK

Q85

Second-most expensive product per category
Find the product with the second-highest distinct price in each category.
DENSE_RANK

LEVEL 12 Date and Monthly Sales Analysis
SQLite date extraction, monthly ranking, running totals and growth calculations.

Q86

Monthly sales summary
Find total order count and total sales for each year-month.
STRFTIME / GROUP BY

Q87

Monthly average order value
Calculate the average order amount for each year-month.
STRFTIME / AVG

Q88

Highest-value order by month
Find the highest-value order in each month.
RANK / PARTITION BY

Q89

Lowest-value order by month
Find the lowest-value order in each month.
RANK / PARTITION BY

Q90

Monthly sales ranking
Rank months based on total sales from highest to lowest.
CTE / RANK

Q91

Running monthly sales total
Calculate cumulative sales across successive months.
SUM OVER

Q92

Previous-month sales
Show each month's sales together with the previous month's sales using LAG().
LAG

Q93

Month-over-month sales change
Calculate the numerical sales difference between each month and the previous month.
LAG / CALCULATION

Q94

Month-over-month growth percentage
Calculate percentage increase or decrease in sales compared with the previous month.
LAG / NULLIF

Q95

Best-performing month
Find the month with the highest total sales.
CTE / MAX

LEVEL 13 Advanced Assessment Questions
Multi-stage analytical reports combining CTEs, aggregation and window functions.

Q96

Customer's most frequently ordered category

Find the product category ordered most frequently by each customer.

MULTI-STAGE CTE

Q97

Customer's highest-spending category

Find the category in which each customer spent the most.

GROUP BY / RANK

Q98

Customers contributing the first 50% of revenue

Rank customers by spending and use cumulative revenue percentage to identify the customers contributing the first 50% of revenue.

RUNNING TOTAL

Q99

Employee department comparison report

For each employee, show salary, department average, difference from average and salary rank within department.

WINDOW FUNCTIONS

Q100

Comprehensive customer analytics report

Create a report with customer details, order metrics, purchase diversity, dates, latest order, spending rank and revenue contribution.

ADVANCED CTEs

Completion checklist

- The query output matches the requested grain.
- Aggregations occur before ranking where required.
- SQLite date functions are used correctly.
- Ties and NULL values are handled deliberately.
- Aliases and CTE names clearly describe their purpose.